

Log Reading Assignment

This assignment consists of two production log files captured over the same time period. Together, they contain all the information needed to investigate a production incident.

Files

web.log – Logs from the web application. Each line represents an incoming HTTP request and includes information such as the request method, endpoint, status code, **user_id**, and **request_id**.

worker.log – Logs from the background worker responsible for processing asynchronous jobs created by the web application. Entries reference only a **request_id**, which can be used to correlate events with **web.log**.

The logs contain a mixture of normal application activity and unrelated errors. Your goal is to identify the events that are actually responsible for the incident.

The Problem

Customer support reported that users were successfully placing orders, but some of those orders never appeared in the system. No monitoring or dashboards were available at the time, so your only source of information is the two log files.

Investigate the logs and answer the questions below.

Questions

1. When did the problem start? Include the timestamp and the evidence you used.
2. Which endpoint is affected? Support your answer with evidence from the logs.
3. What do the failing requests have in common? Identify the pattern and support it with numbers.
4. How many distinct users were affected?

Bonus: Is there anything in the logs that suggests the root cause?

Submit

- **ANSWERS.md** containing your answers, with supporting evidence such as timestamps, counts, or sample log lines.
- Any scripts or code you used to analyze the logs.
- A short section describing your investigation process (5–10 bullet points is sufficient), including anything you investigated that turned out to be unrelated.

Notes

- You may use any language or tools, including AI assistants.
- Do not submit the log files.
- Your submission should reflect your own investigation and analysis.